



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING
BEE 5202: SUSTAINABLE ENERGY SYSTEMS

Campus Building Energy Audit and Renewable Retrofit

You are required to conduct an energy audit of a selected building on campus and develop a renewable retrofit proposal.

All assumptions must be clearly stated and justified.

Scope and Requirements

1. Building Energy Audit

Select a campus building.

Identify and categorize energy loads (lighting, HVAC, laboratory equipment, etc.).

Estimate annual electricity consumption using engineering calculations based on equipment ratings and operating hours.

What are the major energy consumers?

2. Energy Efficiency Assessment

Propose at least 3 energy efficiency improvement measures for the building.

For each measure:

1. Describe the intervention
2. Estimate annual energy savings
3. Estimate implementation cost

4. Calculate payback period

Present total potential reduction in building energy consumption.

3. Renewable Energy System Design

After demand reduction, design 2 viable and distinct renewable energy options for the building.

4. Techno-Economic Analysis

Perform financial evaluation of your proposed systems.

Include:

1. Capital costs
2. O&M costs
3. Annual energy savings
4. Payback period
5. Net Present Value

Recommend the most suitable system.

5. Deliverables

Report

Calculation spreadsheets

Presentation

6. Report Format

Your report must include:

1. Introduction
2. Description of selected building
3. Energy audit results
4. Energy efficiency analysis

5. Renewable system designs
6. Techno-economic evaluation
7. Final recommendation
8. References

Note:

Plagiarism is a serious offence. If detected in any form in coursework and assignments the following will apply:

1. In partial or non-serious cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment will be duly struck off.
2. In serious cases (such as whole duplication of a paper), a zero policy will apply i.e., all offending assignments will be awarded a mark of zero.

The level of seriousness referred to above is at the discretion of the lecturer. Appeals are certainly possible through the relevant channels.